



DHARMSINH DESAI UNIVERSITY, NADIAD
FACULTY OF TECHNOLOGY
FIRST SESSIONAL EXAM

SUBJECT: (IT704) DATA ANALYSIS & INFORMATION EXTRACTION

Examination : B. Tech IT Semester – VII	Seat No. : IT132
Date : 03/08/2023	Day : Thursday
Time : 2:30 PM to 3:45 PM	Max. Marks : 36

INSTRUCTIONS:

1. Figures to the right indicate maximum marks for that question.
2. The symbols used carry their usual meanings.
3. Assume suitable data, if required & mention them clearly.
4. Draw neat sketches wherever necessary.
5. Use of calculator is allowed.

Q.1 Do as directed.

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|-----|---|-----|---|------|
| CO1 | R | (a) | State the difference between enterprise data warehouse and data mart. | [12] |
| CO1 | R | (b) | Explain various types data cube measures by giving at least one example of each. | [2] |
| CO2 | U | (c) | Calculate number of cuboids for 8 dimensions. | [2] |
| CO2 | U | (d) | Prepare the bit map index for "Team" attribute from table 1 given below. | [2] |
| CO3 | A | (e) | Consider the data from table 1 given below and transform the data of "No. of 4's scored by player" using Haar-2 Wavelet Transformation. | [2] |
| CO3 | A | (f) | Consider the data from table 1 and identify whether "No. of 4's scored by player" and "No. of 6's scored by player" are correlated or not? Justify your answer. | [2] |

- Q.2 Attempt Any TWO from the following questions.** [12]
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|-----|---|-----|--|-----|
| CO4 | N | (a) | Why do we use sampling? Provide analyze of various sampling technique in detail with example. | [6] |
| CO4 | N | (b) | Analyze the difference between OLAP and OLTP and provide at least 6 differences between them. | [6] |
| CO4 | N | (c) | Which are data warehouse development approaches? Analyze each approach in detail and draw recommended approach for the same. | [6] |

Q.3 Attempt Any ONE from the following questions. [12]

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|-----|---|-----|--|-----|
| CO6 | C | (a) | Suppose that a data warehouse consists of the four dimensions' date, spectator, location and game, and the two measures count and charge.
i) Draw a star schema diagram for the data warehouse.
ii) Starting with the base cuboid [date, spectator, location, game], what specific OLAP operation should one perform in order to get the year wise aggregated count.
iii) Starting with the base cuboid [date, spectator, location, game], what specific OLAP operation should one perform in order to get the aggregated data (aggregated charge paid by spectator) for game="Table Tennis". | [6] |
| CO5 | E | (b) | Consider data from table-1 given below.
i) Use min-max normalization to transform the age of Virat Kohli and MS Dhoni onto the range [0.0,1.0].
ii) Use Z-score normalization to transform the age of Virat Kohli and MS Dhoni, where standard deviation for age is 10 years.
Comment on which method you would prefer to use for normalization on the given data by providing appropriate reason. | [6] |

OR

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|-----|---|-----|--|------|
| CO6 | C | (a) | Design a snowflake schema diagram to store information about weather observation stations. The weather station has about 1000 probes, which are scattered throughout various land and ocean locations in the region to collect | [12] |
|-----|---|-----|--|------|

basic weather data, including air pressure, temperature, and precipitation at each hour. All data are sent to the central station, which has collected such data for more than 10 years. Consider region_map as one of the measures. Also provide a lattice of cuboids by considering any of the 3 dimensions as a base cuboid.

- CO5 E (b) Discretization the RUN scored by each player using following smoothing [6] technique. Consider data from table-1. Take number of bins as 2
- Equal depth (Frequency) and apply data smoothing by bean mean.
 - Equal depth (Frequency) and apply data smoothing by bean boundary.
 - Equal width (distance) and apply data smoothing by bean mean.
- Comment on which method you would prefer to use for data smoothing, by providing appropriate reason.

Table 1: Cricket Player information

#	Batsman	Team	Age	Run	Ball	4s	6s
1	Ravindra Jadeja	CSK	34	10	5	2	0
2	MS Dhoni	CSK	42	25	10	4	1
3	Hardik Pandya	GT	29	20	11	4	0
4	Shikhar Dhawan	RCB	37	40	29	6	0
5	K.L Rahul	LSG	31	30	12	3	0
6	Rohit Sharma	MI	36	45	13	5	1
7	Virat Kohli	RCB	34	80	44	8	4
8	Surya Kumar Yadav	MI	32	43	25	0	3